

Excitation of Seismic Waves

Chapter 4

Various mechanisms can excite seismic waves, such as faulting and volcanic eruptions. These phenomena that excite seismic waves cannot be described within the framework of a linear elastic body. However, if the source region is localized, the representation theorem learned in the previous chapter shows that the problem of seismic wave excitation can be treated within the framework of a linear elastic theory. In this chapter, we will learn that these phenomena can generally be described by an **equivalent body force** within a linear elastic body. In other words, investigating an excitation source in seismology means estimating its **force system**.

Seismological methods are extremely powerful for discovering the characteristics of a seismic source, but it is important to note that wave propagation only provides information about the **force system**. In order to obtain a physical picture of the excitation source (e.g., amount of slip or volumetric change), it is necessary to reinterpret the **equivalent body force** in terms of physical phenomena. Depending on the situation, multiple interpretations may be possible, and the estimated picture of the source strongly depends on the assumed physical model. Thus, when trying to understand the excitation mechanism from seismological observations, it is important to be aware of this two-tiered logical structure.

The following sections will first explain the concept of an indigenous source to understand what an equivalent body force is. Then, we will discuss the moment tensor.

§4.1 Indigenous source

As long as an event that excites seismic waves does not occur, the Earth remains quiet without oscillating. When an external force (a force from outside the system, in this case, the Earth) acts at some point, seismic waves are excited. The easiest example of such an event is a meteorite impact. The ejection of mass associated with a volcanic eruption is another example that can be approximated by an impulse, provided the ejecta dissipates outside the Earth. In this way, when an

external force is applied, the total momentum of the Earth changes by the amount of the impulse caused by the external force.

In most cases, however, the solid Earth can be approximated as a closed system. Even in such a case, phenomena inside the solid Earth can, of course, excite seismic waves. Earthquakes (faulting) are a typical example. Volcanic processes inside the solid Earth, such as the source of volcanic tremors, can also be cited as examples. Such an excitation source within an isolated solid Earth system is called an indigenous source. Specific physical processes of an indigenous source include thermal stress, phase transitions, faulting, and fluid movement (e.g., the source of volcanic tremors). Since the entire Earth can be regarded as an isolated system, the total momentum and angular momentum of the entire system must be conserved.

For an indigenous source to excite seismic waves, Hooke's law must break down in some part of the Earth's interior (otherwise, it will remain stationary). For example, faulting during an earthquake does not obey linear elasticity in the vicinity of the fault. If we consider an elastic body by analogy to weights and springs, this corresponds to a sudden change in the spring constant, such as a spring breaking. Fluid motion also plays an important role when considering the excitation of volcanic tremors. Let us consider a case where a region that does not obey the linear elastic theory is localized within a volume V . In this case, the phenomena can be roughly divided into two categories.

The first relates to cases in which the exchange of momentum and angular momentum between the region V and its exterior (the linear elastic region) can be ignored. Faulting is an example of this. In this case, momentum and angular momentum are conserved at all times even in the linear elastic region (outside V).

The second relates to cases in which momentum and angular momentum are exchanged between the region V and its exterior (the linear elastic region). A good example is a landslide, where a rigid block slides down a slope. At the beginning and end of the motion, the momentum of the sliding block is zero, but it possesses momentum in between. For momentum to be conserved in the Earth as a whole, the region outside V must receive momentum from the sliding block. That is, an impulse acts on it^{note 1)}.

These properties become important when considering the equivalent body force described in the next section.

§4.2 Equivalent body force and stress

Let us now consider the occurrence of an earthquake. An earthquake is a phenomenon in which a discrepancy (slip) occurs on a fault plane. The rupture phenomenon itself cannot be described within the framework of elastic mechanics. Specifically, since Hooke's law is a physical law that should strictly hold elsewhere, we can say that a phenomenon is occurring whose constitutive law

note 1) For details, see Takei and Kumazawa [1994, 1995].^{(14), (15)}

cannot be described by Hooke's law. Here, let us consider a volume V enclosed by Σ . Outside Σ , we assume that the phenomena can be completely described within the framework of a linear elastic body, and that Hooke's law breaks down only inside Σ .

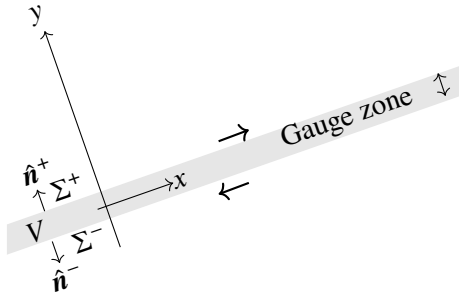


Figure 4.1: Schematic figure of a right-lateral vertical strike-slip fault.

The representation theorem guarantees that if the displacement and stress on Σ are known, the displacement and stress outside of it can be completely described. Information about the deformation inside Σ is not necessary for the deformation outside Σ . Therefore, let us consider virtually embedding a linear elastic body inside Σ as well, which satisfies the boundary conditions of displacement and stress on Σ . In the following, we will show that the entire system can be represented by a linear elastic body, and the effect of the breakdown of Hooke's law can be completely expressed by an equivalent force (**equivalent body force**)^{note 2)}.

Suppose an earthquake (faulting) occurs inside Σ . Inside Σ , brittle failure is occurring, and Hooke's law does not hold. Deformation due to brittle failure occurs in a thin layer known as fault gouge^{note 3)}. Assume we can measure the actual stress and denote it by T_{true} (dashed line in the figure).

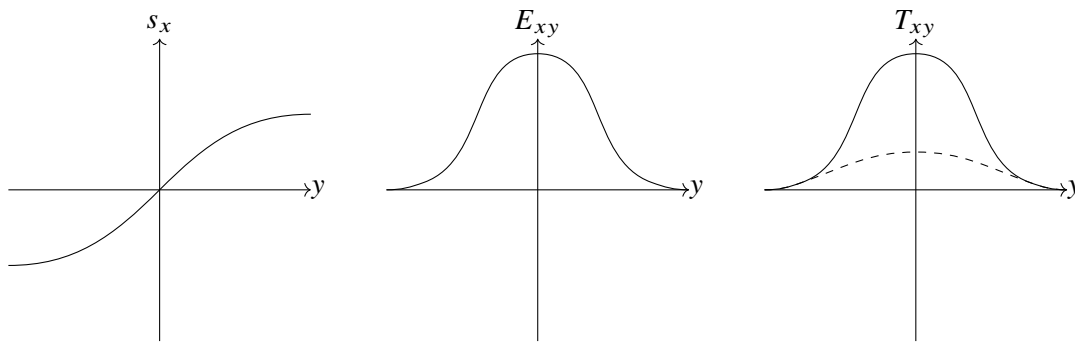


Figure 4.2: Schematic figure of the displacement, strain and stress.

note 2) For details, see Dahlen and Tromp (1998).⁽²⁾

note 3) Gouge consists of unconsolidated particles comminuted by brittle failure. See, for example, the website of Dr. Sakaguchi: <http://www.arito.jp/LecEQ22.shtml>.

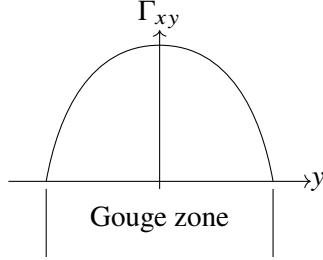


Figure 4.3: Schematic figure of the stress glut.

Here, let us replace the interior of Σ virtually with an elastic body. Since Hooke's law holds inside Σ , we can calculate the stress from the strain. As shown in Figure 4.2, the stress obtained by modeling with a linear elastic body is larger than the actual stress. Let us denote this modeled stress by $\mathbf{T}_{\text{model}}$ (solid line in the figure). We now define the **stress glut** $\mathbf{\Gamma}$ as $\mathbf{\Gamma} = \mathbf{T}_{\text{model}} - \mathbf{T}_{\text{true}}$.⁽¹⁾ On Σ , we have $\mathbf{\Gamma} = 0$.

Let us consider an infinitesimal element within the region replaced by the elastic body. If we apply $-\mathbf{\Gamma}$ inside Σ , the net stress applied to the perfectly elastic body becomes $\mathbf{T}_{\text{true}}$, and its equation of motion matches that before the replacement by the elastic body. In other words, by replacing the region inside Σ with an elastic body and applying $-\mathbf{\Gamma}$ within the region, we can describe the deformation $\mathbf{s}$ within the entire volume V . Assuming the stress glut is distributed inside the elastic body, we can define the equivalent body force as $-\partial_j \Gamma_{ij}$. This allows us to completely describe the problem within the framework of linear elasticity, treating it as a problem where an equivalent body force acts on an elastic body.

Since the solid Earth is a closed system, the net force and torque on the system are zero. In the case of faulting, we can approximate that there is no exchange of angular momentum between the inside and outside of the region Σ , and the equivalent body force can be represented by two pairs of force couples.

When trying to understand a seismic source (the excitation source of seismic waves) from elastic waves, all we can do is determine the excitation region and the equivalent body force. The physical model that can explain the equivalent body force is not necessarily unique, and independent information is essential for considering its excitation mechanism.

Using the stress glut $\mathbf{\Gamma}$, the equivalent body force can be written as

$$\mathbf{f} = -\nabla \cdot \mathbf{\Gamma}(\mathbf{x}, t). \quad (4.1)$$

The excitation of seismic waves by the equivalent body force can be written using Green's functions as

$$\mathbf{S}(\mathbf{x}, \omega) = \int_V \mathbf{G}(\mathbf{x}, \boldsymbol{\xi}, \omega) \mathbf{f}(\boldsymbol{\xi}, \omega) dV(\boldsymbol{\xi}). \quad (4.2)$$

<https://www.eri.u-tokyo.ac.jp/people/knishida/eng/seismology.html>

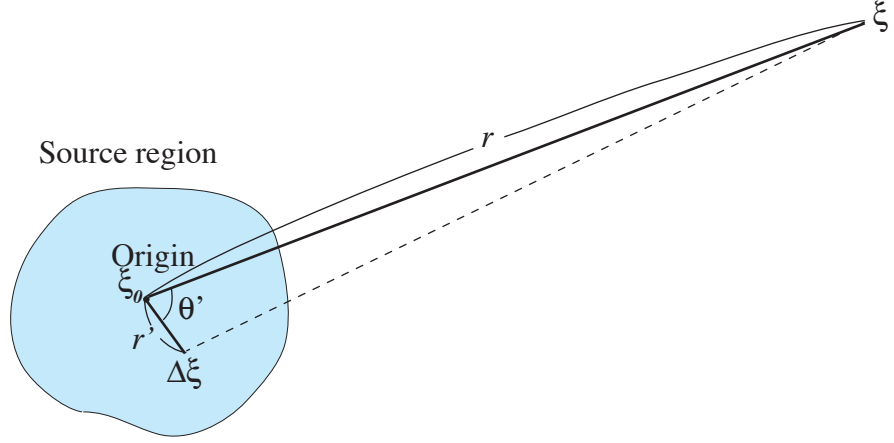


Figure 4.4: Schematic figure of the source-receiver geometry.

§4.3 Multipole expansion

Let us Taylor expand the Green's function with respect to $\Delta\xi$ in the vicinity of ξ_0 . Let k be a typical wavenumber of the Green's function. The n -th derivative of the Green's function G can be evaluated as $k^n G$. Therefore, the n -th order term of the Taylor expansion is approximately

$$\frac{1}{n!} k^n G \Delta\xi^n = \frac{1}{n!} G (k\Delta\xi)^n. \quad (4.3)$$

When $k\Delta\xi$ is sufficiently small, that is, when the dimension of the region Σ is sufficiently small compared to the wavelength of the seismic wave (Figure 4.4), we can apply a Taylor expansion. So, let's substitute the Taylor expansion up to the second-order term with respect to $\Delta\xi$:

$$G(x, \xi, \omega) \approx G(x, \xi_0, \omega) + \nabla_{\xi} G(x, \xi_0, \omega) \Delta\xi + \frac{1}{2} \Delta\xi^T \mathcal{H}_{\xi} G(x, \xi_0, \omega) \Delta\xi + O(\Delta\xi^3). \quad (4.4)$$

Substituting this expression into equation 4.2, we obtain

$$\begin{aligned}
 S_i(\mathbf{x}, \omega) \approx & G_{ij}(\mathbf{x}, \xi_0, \omega) \int_V f_j(\xi, \omega) dV(\xi) \\
 & + \partial_k G_{ij}(\mathbf{x}, \xi_0, \omega) \int_V f_j \Delta \xi_k dV(\xi) \\
 & + \partial_k \partial_l G_{ij}(\mathbf{x}, \xi_0, \omega) \int_V f_j \Delta \xi_k \Delta \xi_l dV(\xi).
 \end{aligned} \tag{4.5}$$

The first term represents the resultant force (single force), and the second term represents the moment tensor. If the change in momentum and angular momentum within Σ can be ignored, the first term becomes zero. In addition, if angular momentum is conserved, the total torque is zero, so the anti-symmetric component of the moment tensor becomes zero. When it is necessary to consider the finite size of the source, the third and higher-order terms also become important. On the other hand, since the 3D Green's function has an r^{-1} dependence far from the source, the n -th term has an r^{-n-1} distance dependence. Thus, higher-order terms attenuate more rapidly. Therefore, at long distances, the equivalent body force can be described by the third term (a quantity called the moment tensor)^{note 4)}.

Let us consider a situation where the excitation source and the other elastic regions do not exchange momentum and angular momentum. The 0th-order term of the multipole expansion corresponds to the impulse, and the 1st-order term corresponds to the torque. Therefore, only the second and higher-order terms have values. Let us look at the second-order term in detail in the next section.

§4.4 Excitation of seismic waves by moment tensor

When the spatial scale of the source region is sufficiently shorter than the wavelength, it can be regarded as a point source, and the stress glut can be written as

$$\Gamma_{ij}(\mathbf{x}, t) = M_{ij}(t) \delta(\mathbf{x} - \mathbf{x}_0). \tag{4.7}$$

^{note 4)}Here we considered the case where the source size is sufficiently small compared to the wavelength. What if the source region is wider? Let us consider a horizontally stratified structure. In this case, the Green's function can be expressed by a sum of Legendre functions (a spherical version of the Fourier series expansion). In this case, the following relation known as the addition theorem holds:

$$\frac{2l+1}{4\pi} P_l(\cos \Theta) = \sum_{m=-l}^l Y_{lm}(\mathbf{x}) Y_{lm}(\xi)^*, \tag{4.6}$$

where Θ is the angle between $\mathbf{x}$ and ξ . If the excitation source is distributed on the Earth's surface, one only needs to evaluate $\int Y_{lm}(\xi) f(\xi) dV$. When the source region is small compared to the wavelength, it corresponds to the Taylor expansion.

<https://www.eri.u-tokyo.ac.jp/people/knishida/eng/seismology.html>

Here, $\mathbf{M}$ is the moment tensor. The trace of the moment tensor M_{ij} represents the volume change. Also, as shown in the previous section, we are considering a situation where there is no exchange of angular momentum inside and outside the source region. Therefore, it is sufficient to define the moment tensor as a symmetric matrix. Since the moment tensor is a symmetric matrix, it can be diagonalized with three eigenvalues and mutually orthogonal eigenvectors. In the case of ordinary earthquakes, the magnitude of two eigenvalues of M_{ij} is much larger than the other one (it can be represented by two pairs of force couples) ^{note 5)}. When the time variation of each moment tensor component is similar, we can write

$$\Gamma_{ij} = \hat{M}_{ij} \delta(\mathbf{x} - \mathbf{x}_0) m(t). \quad (4.8)$$

$m(t)$ is an increasing function normalized such that $\int m(t) dt = 1$. Here, the magnitude of the moment tensor, $\mathcal{M}_0$, is defined as

$$\mathcal{M}_0 = \sqrt{\frac{1}{2} \sum_{i,j} \hat{M}_{ij}^2}, \quad (4.9)$$

and it can be expressed using the unit moment tensor $\hat{\mathbf{M}}$ as

$$\mathbf{M} = \sqrt{2} \mathcal{M}_0 \hat{\mathbf{M}}. \quad (4.10)$$

At sufficiently low frequencies, the moment tensor can be simplified using the Heaviside function $H(t)$ as

$$\Gamma_{ij} = \sqrt{2} \mathcal{M}_0 \hat{M}_{ij} \delta(\mathbf{x} - \mathbf{x}_0) H(t). \quad (4.11)$$

At a sufficiently distant observation point, the displacement of body waves (U) in an infinite homogeneous medium can be written as

$$U \sim \frac{\mathcal{M}_0}{r} \dot{m}(t - r/c), \quad (4.12)$$

where r is the distance from the source. This shows that the displacement waveform of a teleseismic body wave represents the moment rate function $\mathcal{M}_0 \dot{m}(t)$.

Problem 4.1

1. In section 3.4.2, we considered an explosive source, and in this case, the source can be represented by a moment tensor. Derive the expression for the corresponding moment tensor when Δr is sufficiently small.
2. Compute the P-wave radiation pattern when the moment tensor has a value only for $M_{xx} = 1$.
3. Compute the P-wave radiation pattern when the moment tensor has values $M_{xx} = 1, M_{yy} = -1$ (see Figure 4.6).

^{note 5)}For example, explosive sources can also be represented by a moment tensor. For details on non-double couple components, see, for example, Julian et al. [1998].⁽⁶⁾

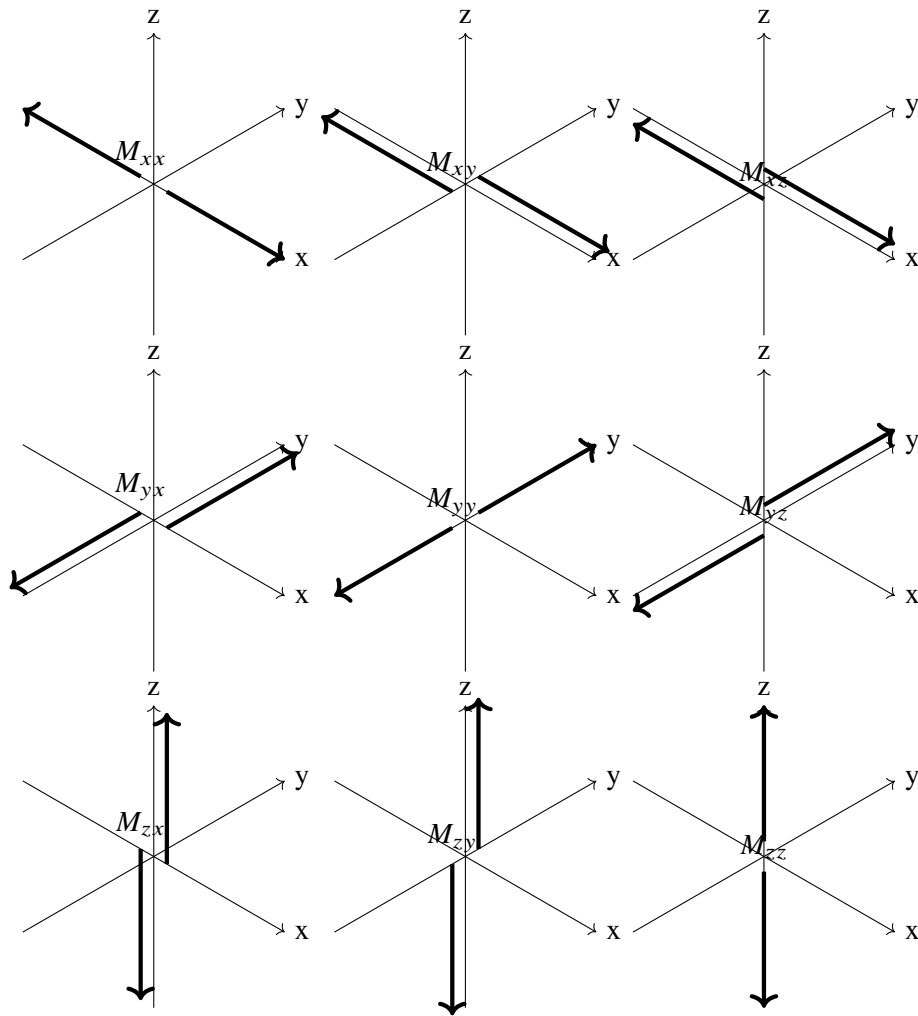
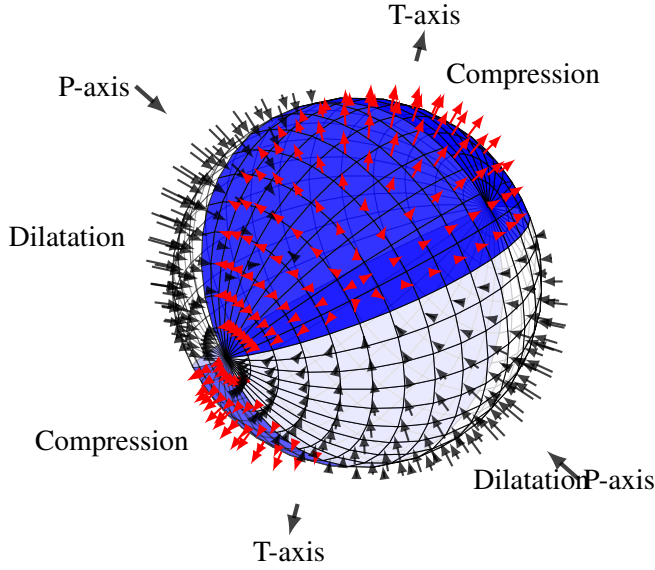


Figure 4.5: Each component of the moment tensor.



Seismic focal mechanism and Pression-Tension axis.

Figure 4.6: Radiation pattern of P-waves by a double couple source. Cyril Langlois (2010)/ CC BY 2.5.

§4.5 Work done by the moment tensor

In this section, let us consider the work done by the moment tensor on an elastic body. First, consider the stress glut Γ , and assume that the corresponding equivalent body force f ,

$$f = -\nabla \cdot \Gamma(x, t), \quad (4.13)$$

acts on the elastic body. Let s be the displacement of the elastic body. The work W done by the equivalent body force on the elastic body is

$$W = \int_V f \cdot s dV. \quad (4.14)$$

Using integration by parts (Problem 4.2), this equation can be rewritten as

$$W = \int_V f \cdot s dV = \sum_{ij} \int_V E_{ij} \Gamma_{ij} dV. \quad (4.15)$$

Here, for simplicity, we assumed that the stress glut is 0 on the surface Σ of the elastic body. When the stress glut can be expressed by a moment tensor as in equation (4.7), it can be simplified to

$$W = \sum_{ij} E_{ij} M_{ij}(t). \quad (4.16)$$

In other words, the work can be expressed as the product of the strain E_{ij} caused by the earthquake at the source and the moment tensor M_{ij} . Considering the law of conservation of energy, the work W done by the stress glut Γ transforms into kinetic energy and elastic energy.

Problem 4.2

Derive equation (4.15) assuming the following conditions:

- Consider an elastic body within a volume V , and assume that the stress glut Γ exists only within a region enclosed by Σ .
- On the surface Σ , $\Gamma_{ij} = 0$.
- Assume the elastic body is an isolated system, and use the condition that the torque acting on the elastic body is zero ($\Gamma_{ij} = \Gamma_{ji}$).
- Hint: Use integration by parts and Gauss's divergence theorem.

§4.6 Effect of free surface on seismic wave excitation

In this section, let us consider how a free surface affects the excitation of seismic waves^(2)note 6). To conclude beforehand, an initially puzzling result is obtained: shallow earthquakes do not excite seismic waves through their M_{xz} and M_{yz} components.

For simplicity, let us consider a semi-infinite medium. Assuming the Earth's surface is at $z = 0$, the free boundary condition is $T_{xz} = T_{yz} = T_{zz} = 0$ at $z = 0$. Rewriting the first two conditions in terms of strain yields

$$E_{xz}|_{z=0} = \frac{T_{xz}}{\mu} \Big|_{z=0} = 0 \quad (4.17)$$

$$E_{yz}|_{z=0} = \frac{T_{yz}}{\mu} \Big|_{z=0} = 0. \quad (4.18)$$

We established that the work W done by the moment tensor is $\sum_{ij} E_{ij} T_{ij}$. Since $E_{xz} = E_{yz} = 0$ at $z = 0$, it is clear that the corresponding moment tensor components M_{xz} and M_{yz} do not contribute to excitation. In other words, near a free surface (where the depth is sufficiently shallow compared to the wavelength of seismic waves), if the hypocenter depth can be considered sufficiently shallow compared to the wavelength, the boundary conditions at the free surface approximately hold for the strain field near the source. Thus, M_{xz} and M_{yz} cannot excite seismic waves. What does this mean in terms of actual physical phenomena?

For instance, shallow earthquakes near trenches often involve massive low-angle thrust faulting. In such cases, the M_{xz} and M_{yz} components do little to excite seismic waves, making it notoriously difficult to determine these components from seismic data. This problem is especially pronounced when estimating the moment tensor using long-period seismic waves. When attempting to find the moment of a shallow, low-angle thrust earthquake, a trade-off arises between the fault dip angle and the seismic moment, resulting in significant uncertainty in the estimated moment.

^{note 6)} See, for example, section 5.4.6 Problematical sources of Dahlen and Tromp [1998]⁽²⁾

§4.7 Excitation of seismic waves by single force

When the excitation source (the region that deviates from linear elasticity) is sufficiently large and its change in momentum cannot be neglected, the term corresponding to the dipole in the multipole expansion cannot be ignored. This term, called a single force, is generated by the exchange of momentum (impulse) between the source region and the surrounding regions. This concept is easier to visualize when considering a landslide (Figure 4.8), for example. Naturally, the total momentum of the entire system must be conserved in such processes.

Examples where a single force becomes important include: (i) landslides, (ii) glacial earthquakes,^{(3), (11)} and (iii) excitation of seismic waves by ocean waves. In all these cases, the motion of a source with a sufficiently large volume corresponds to the excitation of seismic waves.

4.7.1 Volcanic Eruptions

By simplifying the process of a volcanic eruption, we can consider a scenario where material is ejected upward from near the Earth's surface as a plume. In this case, since the plume carries upward momentum, the solid Earth receives a downward single force as a reaction. During the 2022 eruption of the Hunga Tonga-Hunga Ha'apai volcano, seismic waves excited by such a vertical single force were actually observed on a global scale.⁽⁴⁾

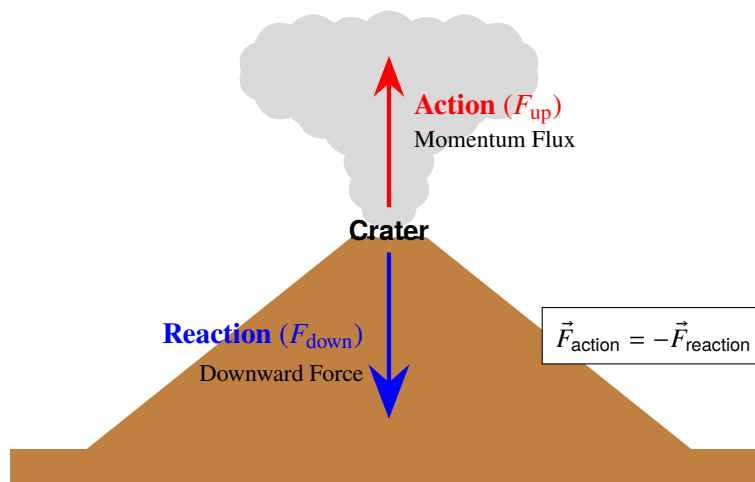


Figure 4.7: Schematic figure of a volcanic eruption.

4.7.2 Landslide

Let's consider the forces at play during a landslide. As illustrated in Fig. 4.8, consider a block resting on a slope. In this state, the component of gravity pulling the block downslope is perfectly balanced by the frictional force. Simultaneously, in the direction perpendicular to the slope, the elastic force (normal force) and gravity are in equilibrium.

If, for some reason, the static friction can no longer support the block, the frictional resistance weakens, and the block accelerates downslope. From the perspective of the solid Earth, because the reaction force acting downslope (which was previously balancing the system) decreases, the Earth experiences a net upward force along the slope.

When the block eventually comes to a halt, it transfers the momentum it gained back to the solid Earth through friction during deceleration. Consequently, the solid Earth experiences a downward force along the slope during this stopping phase.⁽⁷⁾

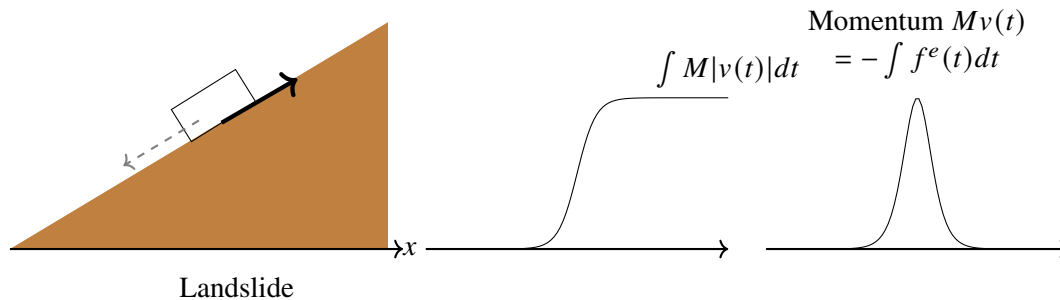


Figure 4.8: Schematic figure of the single force.⁽⁷⁾

Next, let's look at microseisms in more detail.

4.7.3 Microseisms: ocean waves shake the Earth

The Earth is constantly shaking, even when earthquakes are not occurring. In the short period band (around 1 second), human activity is the primary excitation source. At periods longer than 1 second, the vibration originating from human activities diminishes. This is because as the period increases, the wavelength of the seismic waves becomes longer, which means the excitation source needs to shake a deeper (scale on the order of km) region. In the period band longer than 5 seconds, ocean waves are constantly shaking the solid Earth, exciting surface waves (Rayleigh waves and Love waves). This phenomenon is called microseisms.

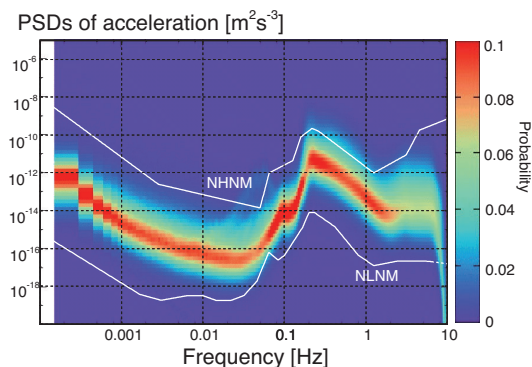


Figure 4.9: Probability density of power spectrum of horizontal acceleration at a Hi-net station in Japan.⁽⁶⁾ A reddish color is more probable. The thick red line represents the power spectral densities of ambient noise.

Let us take a look at Figure 4.9. We can see a peak around the period of 10-20 seconds and a larger peak around the 8-second period. The lower frequency one is called primary microseisms, and the higher one is called secondary microseisms. Primary microseisms corresponds to the frequency of the ocean waves themselves, while secondary microseisms corresponds to double the frequency. Although it seems counterintuitive that the amplitude of the secondary microseisms is larger, this is due to the dominance of nonlinear effects as an excitation mechanism, making harmonics prominent (known as the Longuet-Higgins mechanism: Longuet-Higgins, 1950⁽⁹⁾). The amplitude of secondary microseisms is orders of magnitude larger, and microseisms are clearly observed even in the middle of a continent far from the ocean.

A ripple pattern by raindrops is analogous to those of microseisms. Please run an application at [the website](#). An impulsive force at a point generates an outgoing concentric wave. On the other hand, we can trace wavefront generated by many random sources at first. Gradually, inside the circle, the wave field becomes quite random. We cannot identify any specific direction.

It is observed particularly clearly when a typhoon passes, as the wave height increases. Figure 4.10 shows an example during the passage of Typhoon No. 18 in 2004. A running spectrum^{note 7)} was calculated from approximately one and a half days of records at Minamidaito Island. As the typhoon weakens and moves away from the observation point, the amplitude can be seen to decrease. Also, the dominant period becomes shorter as the strength weakens. Because the arrival time is delayed in proportion to the frequency, the effect of deep-water wave dispersion (see Section 9.1.2) can also be observed.

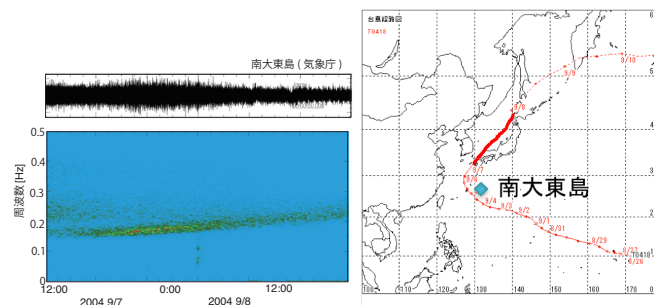


Figure 4.10: Left: Running spectrum from Sept. 7th to Sept. 8th in 2004. The vertical axis shows frequency, and the horizontal one shows time. Right: Track information about the typhoon. The red line shows the track. The thick red line corresponds to the time period of the running spectrum.

A faint vertical streak can be seen in Figure 4.10. This is a seismic wave from a distant earthquake. In this way, microseisms hide seismic wave signals. For seismic observation, microseisms constitute a major source of noise.

note 7) A running spectrum is a figure showing the temporal variation of a power spectrum. It is used to investigate changes in frequency over time. To calculate it practically: (1) cut out data from the original data while shifting the time window, (2) calculate the power spectrum for each segment of cut-out data, and (3) arrange the calculated power spectra in order along the time axis.

Excitation mechanism of microseisms: Longuet-Higgins mechanism

In this section, we will explain the excitation mechanism of microseisms based on Longuet-Higgins [1950]⁽⁹⁾^{note 8)} ^{note 9)}. See Section 9.1.2 for more on water surface waves.

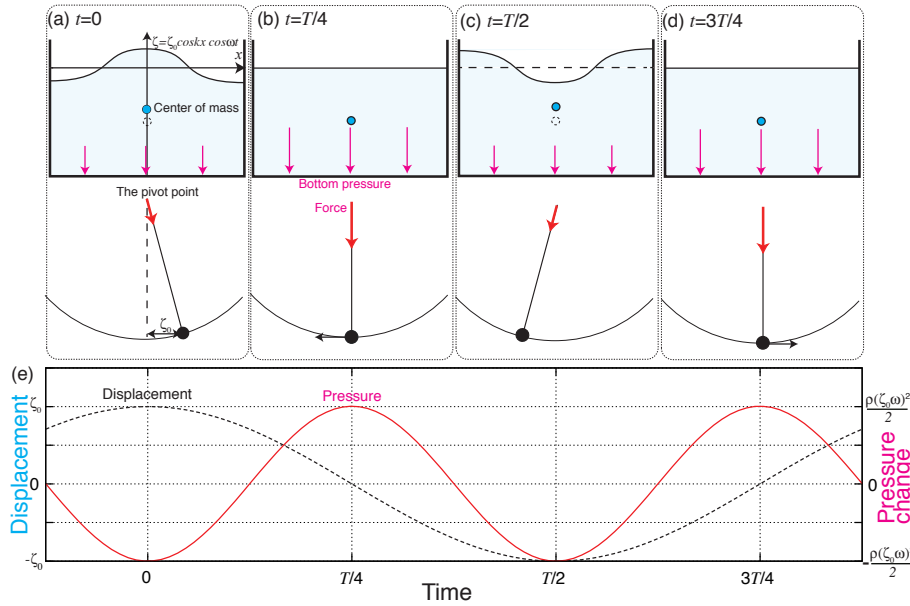


Figure 4.11: Top: Simplified model (a-d) and pendulum analogy. In the top left figure, waves do not propagate because a standing wave is considered. Taken from Nishida (2017).⁽¹³⁾

Let us consider a pendulum as an analogy, as shown on the right of Figure 4.11 (Longuet-Higgins, 1953⁽¹⁰⁾). In the left figure, since a standing wave is considered, the wave does not propagate but moves vertically. Therefore, considering the position of the center of gravity at each instant, it is higher in (a) and (c) than in (b) and (d). To cause an up-and-down movement of the center of gravity, a pressure fluctuation with a frequency of 2ω is necessary. This is easy to understand if we think based on the pendulum on the right side of the figure. The position of the pendulum represents the position of the center of gravity, the force at the fulcrum corresponds to the pressure at the ocean floor, and the horizontal displacement of the pendulum corresponds to the depth of the center of gravity of the seawater. The swinging of the pendulum corresponds to the movement of water (see also the streamlines at the bottom of Figure 4.11). Considering the fluctuation of the force at the fulcrum, one can see that the pressure fluctuates at 2ω . From a simple calculation, it can also be seen that the pressure fluctuation is proportional to the square of the wave amplitude. In other words, this pressure fluctuation is a second-order term (nonlinear effect).

^{note 8)} Longuet-Higgins was an applied mathematician and physical oceanographer who actively conducted various studies on the statistical theory of ocean waves and ocean waves in general.

^{note 9)} Although the handling of the nonlinear term here is somewhat ad hoc, it has been simplified. For details, refer to Longuet-Higgins (1950), Hasselmann (1963)⁽⁵⁾^{note 10)}, and Kedar *et al.* (2008).⁽⁸⁾

Next, let's consider a traveling wave instead of a standing wave. In the case of a traveling wave, it simply moves at a constant speed while preserving its shape, so the center of gravity does not change. That is, no pressure fluctuation occurs. We can see that the Longuet-Higgins mechanism is valid only for standing waves and not for traveling waves. In other words, to excite microseisms by the Longuet-Higgins mechanism, at least one pair of ocean waves traveling in opposite directions is necessary. It is inferred that the reflection of waves at the coast and the existence of multiple wave sources are essential for the excitation of microseisms.

§4.8 Bibliography

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